

# 1 Palletizing and Packing on the AIP

## Log packing OP on

Use the input function (dialog A\_AN\_HU or A\_P\_AN\_HU) to log on a packing OP requiring batch management to the terminal.

When logging the OP on, an operation is selected from the sequencing list and logged on to the machine. The logon is performed without input material for OPs requiring batch management and posted using the PDM command A\_AN or A\_P\_AN.

If the machine is configured so that the person has to be logged on with the OP, the A\_P\_AN command is used and the dialog has to include an input field for the badge number.

## Assign batches

The below input function (dialog CE\_AN\_HU) is used on the terminal to assign batches to the current TPU (running output batch) for a running OP requiring batch management:

**Assign batches** <CE\_AN\_HU>

Machine: MPDV-PAC      Transp. unit: HUPBF9011D  
Order: PLATE000320000850

Batch No.

Batches in transport unit

| * Batch No.  | Material    | Weight | Einh | Designation             |
|--------------|-------------|--------|------|-------------------------|
| - PRPBF90116 | PLATE_WIP07 | 1,00   | KG   | Plate finishing Article |

No. of batches: 1      Total qty.: 1  
Staff badge number:

Figure: "Assign batches" dialog - CE\_AN\_HU

Enter a batch number that is known in HYDRA and click the button *Assign batch* to logically assign the batch to the current TPU. Once assigned successfully, the batch is set to the "processed" status and displayed in the list, which includes all batches assigned to the TPU.

**Note:** You can assign only batches of the same material.

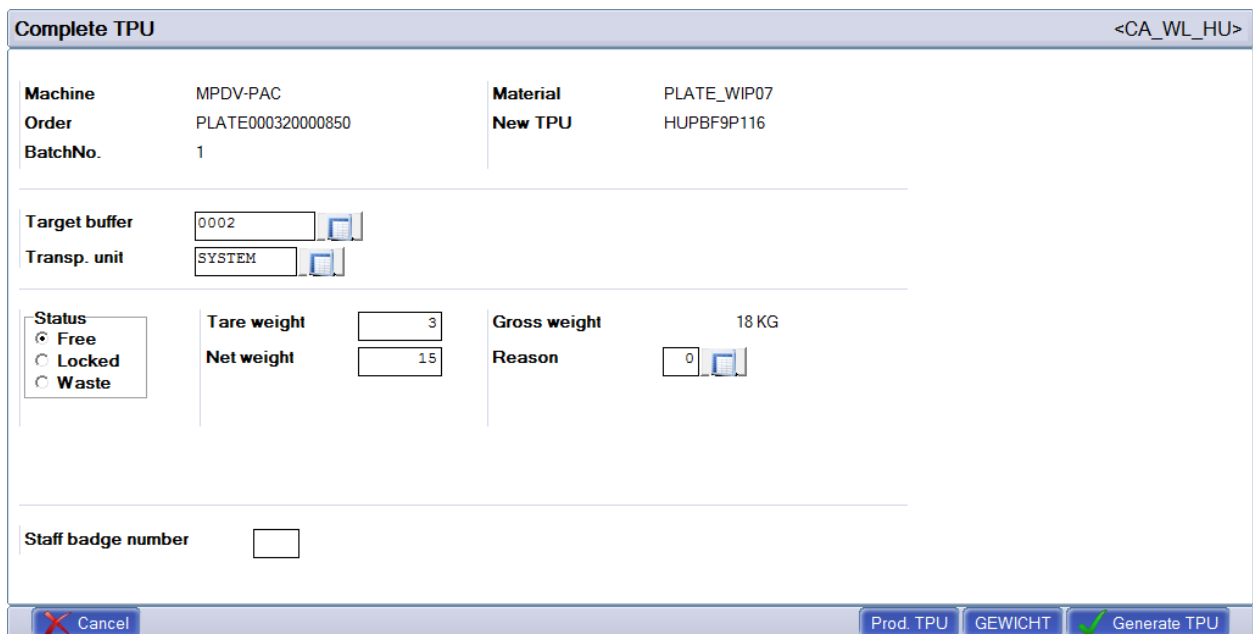
Use the button *Remove batch* to cancel the assignment of a selected batch (double-click on the relevant row). Then the batch changes back to the "free" status.

As additional status information on the current TPU, the dialog shows the number of assigned batches and their total quantity in kg.

You configure the contents of the "batches in transport unit" table in the section [ C\_PAL\_ZUORD.LST ] of the file ctaipplay.ini.

## Complete TPU

You use this input function (dialog CA\_WL\_HU) to complete the currently running output batch (TPU) on the terminal.



|                                                                                                                       |                                              |                                              |  |
|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------|----------------------------------------------|--|
| <b>Machine</b> MPDV-PAC                                                                                               |                                              | <b>Material</b> PLATE_WIP07                  |  |
| <b>Order</b> PLATE000320000850                                                                                        |                                              | <b>New TPU</b> HUPBF9P116                    |  |
| <b>BatchNo.</b> 1                                                                                                     |                                              |                                              |  |
| <b>Target buffer</b> 0002                                                                                             |                                              |                                              |  |
| <b>Transp. unit</b> SYSTEM                                                                                            |                                              |                                              |  |
| <b>Status</b><br><input checked="" type="radio"/> Free<br><input type="radio"/> Locked<br><input type="radio"/> Waste | <b>Tare weight</b> 3<br><b>Net weight</b> 15 | <b>Gross weight</b> 18 KG<br><b>Reason</b> 0 |  |
| <b>Staff badge number</b>                                                                                             |                                              |                                              |  |

Buttons: Cancel, Prod. TPU, GEWICHT, Generate TPU

Figure: "Complete TPU" dialog – CA\_WL\_HU

Using the function *Complete TPU*, you can complete a TPU and enter the weight and the status yield, locked, scrap. When you complete the TPU, the relevant quantity (net weight) is booked to the running output batch and completed and a goods receipt (movement type 101) is posted. At the same time, the next output batch is logged on to the OP.

You can collect the following data with this posting:

### Target buffer

Material buffer where the TPU batch is posted. By default, this is the machine's output buffer.

### Gross weight/tare weight/net weight

By default, the total (in kg) of the assigned single batches is entered as net weight. If entered manually, it is calculated as follows:

$$\text{Gross weight} = \text{Tare weight} + \text{Net weight}$$

The values "gross weight" and "tare weight" are additionally saved as batch attributes for the TPU batch:

Gross weight - batch attribute 301 (ATTR:301)

Tare weight - batch attribute 302 (ATTR:302)

### **Transp. unit (transport unit)**

A transport unit existing in the system can be assigned to the TPU batch.

**Note:** the weight stored for the transport unit is currently not taken over as tare weight to the input dialog.

### **Status (free/locked/waste)**

By using the status selection, you can assign a status to the generated TPU batch: status "free" (yield material), status "locked" (class: yield material) or status "waste" (class: scrap).

In case of scrap, the quantity is posted as scrap for the operation and a scrap reason is entered.

### **Number of batches**

Shows the number of batches currently assigned to the TPU. The value is saved as batch attribute 201 (ATTR:201).

Once the input has been confirmed by clicking OK, you can optionally enter further batch attributes if the attributes have been defined for the material type of the operation.

Once the posting has been completed successfully, the TPU batch is generated as "merged batch" and the relevant quantity and the goods movement are posted for the operation.

## **Log OP off / Interrupt OP**

The packing OP can be interrupted or logged off on the terminal using this input function (dialog A\_UN\_HU or A\_AB\_HU).

Once selected, the user can interrupt or finish the packing OP:

Interrupt OP

<A\_UN\_HU>

Workplace

MPDV-PAC

Operation

PLATE000320000850

Material

PLATE\_WIP07

---

Status

1

---

Staff badge number

☐ Memorize

Cancel

Interrupt OP

Figure: "Interrupt packing OP" dialog – A\_UN\_HU

The running packing OP is interrupted/finished after confirming by clicking OK.

Notes / restrictions:

- The last active output batch of the packing OP is completed without quantity and with status "deleted".
- If batches are still assigned to the TPU when you try to log the OP off, the logoff is rejected with error message.
- The ADE checks for overdelivery/underdelivery of the operation are not active when you use these posting functions.
- Quantities cannot be posted using this function.

## Unpack / Repack

### Requirements

Service pack 13 must be installed and activated. The document [Activating MPL TRT Dialogs \(SP13\)](#) describes how to use the dialog on the shop floor client.



For new customers, the new/updated MPL or TRT dialogs are directly available once the service pack 13 has been released.

## General

Using this dialog, the operator can enter the packed handling unit and unpack selected batches. The operator can also unpack all batches and therefore cancel the handling unit.

### Terminal procedure - Unpack individual batches

You start the function "Unpack TPU" (Dialog: CE\_DEL\_HU) in the main view by clicking the button *Unpack TPU*.

The user enters the handling unit or merged batch number. The number can be entered manually or by scanning the barcode. If you enter the number manually, you must click the green arrow. If you scan a barcode, the system automatically performs the transfer and requests the detail data of the batch.

The list displays the data of the entered handling unit. You can double-click specific batches to select the batches that you want to unpack. The selected batches are labeled in the list with an "x". Double-click a selected batch to deselect the batch. The display returns to normal ("-").

Use the button *Unpack selected ones* to unpack selected batches.

A confirmation prompt opens asking whether you want to execute the dialog:

- Click *No* to cancel execution. The user returns to the dialog. The selected batches remain.
- Click *OK* to execute the dialog.

Then a message confirms that the unpacking process has been completed successfully. If you click *OK*, you get back to the dialog for further processing.

If you click *Exit*, the dialog closes. Selected unpacked batches remain unchanged in the handling unit.

### Terminal procedure - Unpack all batches

You start the function "Unpack TPU" (Dialog: CE\_DEL\_HU) in the main view by clicking the button *Unpack TPU*.

The user enters the handling unit or merged batch number. The number can be entered manually or by scanning the barcode. If you enter the number manually, you must click the green arrow. If you scan a barcode, the system automatically performs the transfer and requests the detail data of the batch.

The list displays the data of the entered handling unit. You can double-click specific batches to select the batches that you want to unpack. The selected batches are labeled in the list with an "x". Double-click a selected batch to deselect the batch. The display returns to normal ("-").

The function key *Unpack all* unpacks all batches contained in the handling unit.

Then a message confirms that the unpacking process has been completed successfully. If you click *OK*, the dialog is closed because there are no more batches.

## Terminal procedure - Add new batch

You start the function "Unpack TPU" (Dialog: CE\_DEL\_HU) in the main view by clicking the button *Unpack TPU*.

The user enters the handling unit or merged batch number. The number can be entered manually or by scanning the barcode. If you enter the number manually, you must click the green arrow. If you scan a barcode, the system automatically performs the transfer and requests the detail data of the batch.

The list displays the data of the entered handling unit.

If you click the button *Add*, you can add batches to the handling unit. A dialog opens where you can enter the batch number that is added, either manually or by scanning. After confirmation, the batch is added to the handling unit.

## Posting procedure - Unpack individual batches

If you want to remove child batches from a handling unit, the system checks the following:

- The child batches must exist, but can be archived.
- The handling unit must be available but not running or processed.

The following actions are performed for each unpacked child batch:

- The batch status of the child batch is set to the status of the handling unit.
- The remaining quantity of the child batch is calculated from the total quantity of the goods movements when the child batch is packed.
- An entry in the [Batch history](#) is generated.
- The connection between handling unit and child batch is removed
- The batch assignment between child batch and handling unit is removed.

Actions performed for the handling unit:

- The remaining quantity of the handling unit is reduced by the sum of all remaining quantities of all unpacked batches.

- If the remaining quantity of the handling unit is zero, the batch is set to status "A" (processed)
- A goods movement is generated specifying the reduced quantity.

## **Posting procedure - Unpack all batches**

See above "Unpack individual batches". The same processing is applied. All batches are unpacked and the handling unit with the remaining quantity = 0 is set to status "A" (processed).

## **Posting procedure - Add new batch**

If you want to add child batches to an existing handling unit, the system checks the following:

- The handling unit must be available but not running or processed.
- The batch to be added must have the status "F" (free) and the same article number as the handling unit batch.

Note for the added child batch:

- All batch data of the handling unit is transferred to the child batch.
- The batch class of the child batch is identical to the batch class of the handling unit.
- The batch status is "A" (processed).
- The child batch is linked to the handling unit.
- An event „CE\_AN\_PA“ is added for the child batch in the batch history.
- The batch assignment between child batch and handling unit is generated.
- A goods movement (goods issue) is generated for the child batch.

Actions performed for the handling unit:

- The remaining quantity and the quantity of the handling unit increase by the remaining quantity of the added child batch.
- The system generates a goods movement (goods receipt) with the new quantity of the handling unit.